Tensor products and multi-qubit states

Instructions. Show your working. Notation as in the lecture: $|0\rangle = \mathbf{e}_1 = (1,0)^T$, $|1\rangle = \mathbf{e}_2 = (0,1)^T$, $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$, and $|ab\rangle = |a\rangle \otimes |b\rangle$ (similarly for three qubits). For $\mathbf{u} \in \mathbb{C}^2$: $\mathbf{u} \otimes \mathbf{v} = \begin{pmatrix} u_1\mathbf{v} \\ u_2\mathbf{v} \end{pmatrix}$. $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_j \overline{u_j} v_j$. $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and CNOT maps $|a\rangle\,|b\rangle \mapsto |a\rangle\,|b \oplus a\rangle$ (first qubit is the control). Bell states: $\mathbf{\Phi}^{\pm} = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$, $\mathbf{\Psi}^{\pm} = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$.

Question 1

Compute the following tensor products as column vectors.

(a) $\begin{pmatrix} 1 \\ i \end{pmatrix} \otimes \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ i \end{pmatrix}$. Are they equal?

(b) $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ (a vector in $\mathbb{C}^6$).

(c) $\mathbf{e}_1 \otimes \mathbf{e}_2 \otimes \mathbf{e}_2 \in \mathbb{C}^8$. Which basis ket $|abc\rangle$ is it, and in which position is its single 1?

Question 2

(a) Expand $|0\rangle \otimes |-\rangle$ and $|-\rangle \otimes |+\rangle$ as combinations of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

(b) Write $\frac{1}{2}\big(|00\rangle - |01\rangle + |10\rangle - |11\rangle\big)$ as a tensor product of two single-qubit states.

(c) Expand $|1\rangle \otimes |+\rangle \otimes |0\rangle$ as a combination of three-qubit basis kets.

(d) Expand $|+\rangle \otimes |+\rangle \otimes |+\rangle$. How many terms are there, and what is each amplitude?

Question 3

Let $\mathbf{u} = \frac{1}{\sqrt{2}}(1,i)^T$, $\mathbf{v} = \frac{1}{\sqrt{2}}(1,1)^T$, $\mathbf{u}' = \mathbf{v}' = (1,0)^T$.

(a) Compute $\langle \mathbf{u} \otimes \mathbf{v},\, \mathbf{u}' \otimes \mathbf{v}' \rangle$ directly in $\mathbb{C}^4$, and also as $\langle \mathbf{u}, \mathbf{u}' \rangle \, \langle \mathbf{v}, \mathbf{v}' \rangle$.

(b) Show that $\|\mathbf{u} \otimes \mathbf{v}\| = 1$ without writing out $\mathbf{u} \otimes \mathbf{v}$.

(c) Using "inner products multiply", show that $|+\rangle\,|0\rangle$, $|+\rangle\,|1\rangle$, $|-\rangle\,|0\rangle$, $|-\rangle\,|1\rangle$ form an orthonormal basis of $\mathbb{C}^4$.

Question 4

For each two-qubit state $(a,b,c,d)^T$, decide whether it is a product state or entangled. If it is a product, write it as $\mathbf{u} \otimes \mathbf{v}$.

(a) $\frac{1}{2}(1,\ -1,\ 1,\ -1)^T$

(b) $\frac{1}{\sqrt{2}}(1,\ 0,\ 0,\ -1)^T$

(c) $\frac{1}{\sqrt{3}}(1,\ 1,\ 1,\ 0)^T$

(d) $\frac{1}{2}(1,\ i,\ -i,\ 1)^T$

(e) $\frac{1}{\sqrt{5}}(1,\ 2,\ 0,\ 0)^T$

Question 5

(a) For which $\alpha \in \mathbb{C}$ is the (unnormalised) vector $(1,\ \alpha,\ 2,\ 2)^T$ a tensor product? For that α, write it as $\mathbf{u} \otimes \mathbf{v}$ and normalise it.

(b) For which $\alpha \in \mathbb{C}$ is $(1,\ \alpha,\ \alpha,\ 1)^T$ a tensor product? Factor it in each case.

Question 6

(a) Show that the four Bell states form an orthonormal basis of $\mathbb{C}^4$.

(b) Write $|00\rangle$ and $|11\rangle$ as combinations of $\mathbf{\Phi}^+$ and $\mathbf{\Phi}^-$.

(c) Show that $\mathbf{\Phi}^+$ is entangled *without* the $ad-bc$ test: assume $\mathbf{\Phi}^+ = \mathbf{u}\otimes\mathbf{v}$ and reach a contradiction.

(d) Show that $\mathbf{\Psi}^- = \frac{1}{\sqrt{2}}\big(|-\rangle\,|+\rangle - |+\rangle\,|-\rangle\big)$.

Question 7

(a) Write $Z \otimes I,\ \ I \otimes Z$ and $X \otimes X$ as 4×4 matrices.

(b) Compute $(X \otimes X)\mathbf{\Phi}^+,\ \ (X \otimes X)\mathbf{\Psi}^-$ and $(Z \otimes Z)\mathbf{\Phi}^+$.

(c) Describe in words what $Z \otimes I$ and $I \otimes Z$ do to the basis kets $|ab\rangle$.

Question 8

Use the mixed-product rule $(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) = A\mathbf{u} \otimes B\mathbf{v}$ and $(A \otimes B)(C \otimes D) = AC \otimes BD$.

(a) Compute $(H \otimes X)\,|00\rangle$ using the rule, and check it against the 4×4 matrix $H \otimes X$.

(b) Simplify $(H \otimes I)(H \otimes X)$.

(c) Show that $H \otimes H$ is unitary without writing out its 4×4 matrix.

(d) Compute $(H \otimes H \otimes H)\,|100\rangle$ and list the sign of each of the 8 basis kets.

Question 9

(a) Compute CNOT applied to $|+\rangle\,|0\rangle,\ \ |+\rangle\,|1\rangle,\ \ |-\rangle\,|0\rangle,\ \ |-\rangle\,|1\rangle$. Which states do you get?

(b) Compute CNOT applied to $|+\rangle\,|-\rangle$. Is the result a product or entangled? Which qubit changed?

(c) Suppose CNOT $= A \otimes B$ for some 2×2 matrices A, B. Using (a), explain why this is impossible.

Question 10

Measuring only the first qubit. Let $P_0 = \mathbf{e}_1\mathbf{e}_1^\dagger$, $P_1 = \mathbf{e}_2\mathbf{e}_2^\dagger$, and let

$$\boldsymbol{\psi} = \alpha_{00}\,|00\rangle + \alpha_{01}\,|01\rangle + \alpha_{10}\,|10\rangle + \alpha_{11}\,|11\rangle\,, \qquad \sum |\alpha_{ab}|^2 = 1.$$

Recall that measuring *both* qubits gives the outcome ab with probability $|\alpha_{ab}|^2$.

(a) Show that $(P_0 \otimes I)\boldsymbol{\psi} = \alpha_{00}\,|00\rangle + \alpha_{01}\,|01\rangle$, and find $(P_1 \otimes I)\boldsymbol{\psi}$ similarly. In words: what does $P_a \otimes I$ keep, and what does it remove?

(b) Show that $P_a \otimes I$ is a projector (Hermitian, and equal to its own square), and that $(P_0 \otimes I) + (P_1 \otimes I) = I$ and $(P_0 \otimes I)(P_1 \otimes I) = 0$.

(c) Suppose we measure both qubits but only look at the first bit. Then $\Pr(\text{first} = a) = \sum_b |\alpha_{ab}|^2$. Show that this equals $\|(P_a \otimes I)\boldsymbol{\psi}\|^2$.

(d) After seeing first bit a, the two qubits are left in $(P_a \otimes I)\boldsymbol{\psi}$, renormalised to length 1. For $\boldsymbol{\psi} = \frac{1}{\sqrt{6}}\big(|00\rangle + |01\rangle + 2\,|10\rangle\big)$, find both probabilities and both resulting states. Which operator would you use to measure only the *second* qubit?

(e) Measure the first qubit of $\boldsymbol{\Phi}^+$, and of $|+\rangle\,|+\rangle$, and suppose the outcome is 0 in each case. What is then the probability that the second qubit gives 0?

(f) Three qubits: measure the first qubit of $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ using $P_a \otimes I \otimes I$. What are the possible outcomes, and what happens to the other two qubits?